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E. Edge Queries

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given a **directed unweighted** graph with N nodes and M edges. The nodes are numbered from 1 to N . Your task is to find the difference of indegree and outdegree of each node in the graph.

Input

The first line contains two integers N and M ($1 \leq N \leq 2 \times 10^5, 0 \leq M \leq 3 \times 10^5$) — the number of vertices and the total number of edges.

The second line contains M integers $u_1, u_2, u_3 \dots u_m$ ($1 \leq u_i \leq N$) — where the i -th integer represents the node that is one endpoint of the i -th edge.

The third line contains M integers $v_1, v_2, v_3 \dots v_m$ ($1 \leq v_i \leq N$) — where the i -th integer represents the node that is other endpoint of the i -th edge.

The i -th edge of this graph is from the i -th node in the second line to the i -th node in the third line.

Output

Output a single line with N space-separated integers, where the i -th integer is the difference of indegree and outdegree of node i .

Examples

input	Copy
5 10 2 5 4 3 2 4 3 4 1 3 5 1 5 5 1 2 2 1 3 4	
output	Copy
2 0 -2 -2 2	
input	Copy
5 4 5 3 3 2 1 1 2 4	
output	Copy
2 0 -2 1 -1	
input	Copy
8 7 7 7 7 2 1 4 1 2 6 3 4 2 8 5	
output	Copy
-2 1 1 0 1 1 -3 1	

CSE221 Assignment 04 Summer 2026

Contest is running

17:25:12

Contestant



→ Submit?

Language: Java 21 64bit

Choose file: [Choose File](#) No file chosen

Success!



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